

Coursework Submission Coversheet

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In my ontology, I focus on the basic structural definition of a sandwich from the provided Wikipedia link in the assessment brief, which is a food item made with two slices of bread and a filling. I use this as the base definition of the class Sandwich. Later on, I evolved this model to represent a traditional sandwich from my Mexican roots, the Torta.

The key aspects I focus on are Bread and Filling. From Bread exactly two slices, which distinguishes a sandwich from toast (less) or a club sandwich (more). From Filling is required consisting of cheese, meat or vegetables, which are fundamental sandwich contents

After defining the basic sandwich class, I expanded the ontology with Torta as a subclass, adding optional elements: sauce and spiciness. These properties link to the general Ingredient class, although specific sauce or chilies are not yet defined. This is logically consistent, as the properties constrain their range without requiring detailed classes or individuals enabling incremental modeling. This approach allows me to represent Torta's culinary traits while preserving the sandwich structure.

My class structure is the following:

- Food: The most general class, encompassing all edible items.
 - Ingredient: Represents any component used in food composition.
 - Bread
 - Filling:
 - Cheese
 - Meat
 - Vegetable
 - Sandwich: A type of food structurally defined by bread and filling.
 - ClassicSandwich: A defined class, exactly 2 breads and at least 1 filling.
 - Torta: A subclass of Sandwich, with the optional addition of sauce and spiciness.

Having this structure allows me to properly set all the necessary elements which belong to my Food class separated from the actual final product after the combination of elements through

different axioms. From which it logically makes sense. This hierarchy allows for specialization while keeping the structure intuitive. For example, Torta inherits all sandwich constraints but adds cultural variation without violating the general definition.

My properties structure is the following:

- hasBread
 - Connects a sandwich to its bread components. This property is functional, meaning a sandwich has bread.
- hasFilling
 - Links sandwiches to their internal content. It is asymmetric and irreflexive, ensuring clear, one-way (a sandwich has a filling, not vice versa).
- hasSauce
 - A functional property, because it uses one kind of sauce. It links to Ingredient, is restricted only to Torta, and ensures any sauce must be an ingredient.
- hasSpicy
 - A functional property, because it uses one kind of chili. This property checks the presence of spiciness in certain sandwiches. It is functional, applies only to Torta, and is restricted to Ingredient as well.
- isMadeOf
 - A general-purpose, abstract property used for reasoning over all compositional parts. It is not used directly in axioms but serves as a semantic grouping for all the above properties.

Later I used 2 axioms, the first for the ClassicSandwich class which later is used as a blueprint logically to iterate to Torta class, which I could define both as:

- Classic Sandwich Axiom:
 - A sandwich is made of exactly two slices of bread and is filled with at least one ingredient that is either cheese, meat, or vegetable.

This axiom captures the minimal, essential structure of a sandwich, mirroring common definitions. It is used to define ClassicSandwich, a formally equivalent class to Sandwich.

- Torta Axiom:

- A Torta is a type of sandwich that may additionally contain sauces and spicy ingredients.

This definition uses `only` rather than `some` to indicate optional use of sauces and spicy elements i.e., if they are present, they must be ingredients, but they do not have to be present.

Though any of the axioms I do not explicitly mention `isMadeOf`, all the properties used (`hasBread`, `hasFilling`, etc.) are sub properties of `isMadeOf`.

Some advantages of my approach are the following points:

1. **Clarity and Precision:** Clear structural constraints allow for unambiguous reasoning about sandwich makeup.
2. **Extensibility:** The class and property structure can be expanded to include new sandwich types, ingredients, preparation methods, or dietary restrictions.
3. **Modular Design:** Clean separation of components (bread, fillings, sauces, spiciness) facilitates reuse in broader food ontologies.
4. **Reasoning Support:** Logical property characteristics support automatic inference and classification by ontology reasoners.

At last, some disadvantages of my approach are the following points:

- **Limited Scope:** The focus on structural composition excludes other types of similar foods such as wraps or flatbreads, which may be considered sandwich-like but do not fit the exact bread-and-filling definition used here
- **Lack of Ingredient Individuals:** While ingredient categories exist, individual ingredients (e.g., cheddar, jalapeño) are not yet modeled.
- **Scalability Complexity:** Adding preparation steps, serving contexts, or dietary annotations would require significant ontology growth.